

INSTITUTE OF ACCOUNTANCY ARUSHA



DEPARTMENT OF COMPUTER SCIENCE AND MATHEMATICS

FIELD ATTACHMENT REPORT

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I would like to express my sincere gratitude to Almighty God for giving me strength, health and the ability to complete my field attachment successfully.

I am deeply grateful to the Institute of Accountancy Arusha (IAA) for providing me with the academic foundation and opportunity to participate in field attachment training. The knowledge acquired in the classroom provided an important foundation for understanding and performing various practical tasks during the training.

I would also like to sincerely thank YanaCorp for providing me with an opportunity to undertake practical training within a professional Information and Communication Technology environment. The exposure to cybersecurity, system administration, networking, web application security and security assessment tools significantly contributed to my practical knowledge and professional development.

I extend my appreciation to my field supervisor Mr. Elisha Alex Mkumbo. Technical staff and other members of the organization who provided guidance, technical explanations and assistance whenever I encountered challenges.

Special appreciation goes to my lecturers and college supervisor Mrs. Nancy Swai for their academic guidance and support throughout the field attachment period.

ABSTRACT

This report presents the knowledge, practical skills, and professional experience acquired during my field attachment at YanaCorp, an Information and Communication Technology solutions provider operating in Tanzania.

The field attachment provided an opportunity to bridge the gap between theoretical knowledge acquired during my Bachelor of Cyber Security studies and practical cybersecurity and IT operations in a real-world working environment. During the training period, I participated in various technical activities involving system administration, network security, vulnerability assessment, web application security, and security testing.

Practical activities included configuring Windows Server 2019, Active Directory Domain Services (AD DS), managing Organizational Units (OUs), user accounts, and joining Windows 11 client nodes to the domain. I also administered Ubuntu Server environments, configuring static networking, Nginx web server, and MariaDB database services. Additionally, I worked with digital certificates (normal and wildcard) and SSH key-based authentication for secure remote access.

Regarding web application security, I utilized Burp Suite and FoxyProxy to intercept and analyze HTTP traffic, identifying vulnerabilities such as Path Traversal, Access Control flaws, Authentication issues, and Server-Side Request Forgery (SSRF). I also gained hands-on experience in vulnerability assessment using the Metasploit Framework and Greenbone Vulnerability Management (GVM/OpenVAS) on Kali Linux.

Major challenges: Encountered during the attachment included troubleshooting PostgreSQL database collation errors during GVM installation and managing network configuration conflicts in virtualized lab environments.

In conclusion: The field attachment successfully strengthened my practical capabilities in network security, system administration, and vulnerability assessment, achieving the core objectives of the academic program.

Recommendation: It is recommended that future field training programs allow students more direct exposure to live enterprise security environments and that academic institutions expand practical hands-on labs prior to field placement.

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LIST OF ACRONYMS

AD	-	Active Directory
AD CS	-	Active Directory Certificate Services
API	-	Application Programming Interface
DNS	-	Domain Name System
FTP	-	File Transfer Protocol
GVM	-	Greenbone Vulnerability Management
HTTP	-	Hypertext Transfer Protocol
HTTPS	-	Hypertext Transfer Protocol Secure
IP	-	Internet Protocol
IAA	-	Institute of Accountancy Arusha
SSH	-	Secure Shell
OS	-	Operating System
OU	-	Organizational Unit
RCE	-	Remote Code Execution
SSRF	-	Server-Side Request Forgery
SSL	-	Secure Sockets Layer
TLS	-	Transport Layer Security
URL	-	Uniform Resource Locator
VM	-	Virtual Machine
SIEM	-	Security Information and Event Management

CHAPTER ONE

1.0 INTRODUCTION

Field attachment is an important component of university education because it provides students with an opportunity to apply theoretical knowledge acquired in the classroom to practical working environments.

As a Bachelor of Cyber Security student, the field attachment provided an opportunity to gain practical exposure to cybersecurity operations, system administration, networking, vulnerability assessment and web application security.

During my field attachment at YanaCorp, I participated in different technical activities involving the configuration and administration of computer systems, server environments, network services and cybersecurity tools.

The training also involved practical security testing using tools such as Burp Suite, Metasploit Framework, Wazuh and Greenbone Vulnerability Management (GVM/OpenVAS). These activities provided an opportunity to understand how cybersecurity principles are applied in practical environments.

The structure of this report follows the major sections used in the field attachment report sample, including organizational background, activities performed, challenges, lessons learned, conclusion and recommendations.

1.1 BACKGROUND OF THE ORGANIZATION

YanaCorp is an Information and Communication Technology (ICT) solutions provider that provides technology services to organizations and businesses in Tanzania and across Africa.

According to its current public website, YanaCorp provides services in areas including cybersecurity, IT infrastructure, web hosting, domain registration, software development, automation systems, enterprise ICT equipment and technical training. Its cybersecurity services include vulnerability assessments, penetration testing, compliance support, security hardening and managed security services.

The company also provides IT infrastructure services involving servers, core switching networks, firewalls and storage solutions. Its software and business automation services include systems for areas such as customer relationship management, point-of-sale operations, human resource management, document management, healthcare management and analytics.

YanaCorp also operates an internship and early-career programme that exposes students and recent graduates to practical areas including software development, web development, IT support and infrastructure, cybersecurity.

During my field attachment, my practical learning was mainly related to cybersecurity, system administration, networking and security testing. The practical environment enabled me to work with both Windows and Linux systems and to use cybersecurity tools in controlled laboratory environments.

The organization currently lists its Dar es Salaam office at Sinza A, Mori Street, House No. 12, Ubungo, Dar es Salaam, Tanzania and also lists a Zanzibar office at Mbweni.

1.2 TYPE OF ORGANIZATION

YanaCorp operates as an Information and Communication Technology solutions and services organization.

Its publicly described services cover several technology areas including cybersecurity, IT infrastructure, software development, web hosting, domain registration, business automation and technical training.

From a cybersecurity perspective, the organization provides services associated with protecting information systems and digital infrastructure, including:

- i. Security hardening
- ii. Cybersecurity assessments
- iii. Compliance support
- iv. Managed security services
- v. IT infrastructure security

These services create an environment where cybersecurity students can gain practical exposure to different areas of information security.

1.3 MAJOR SERVICES

Based on the organization's publicly available information, YanaCorp provides several major technology services.

1.3.1 Cybersecurity Services

YanaCorp provides cybersecurity-related services including vulnerability scans, penetration testing, compliance support, security hardening and managed security services.

These services are relevant to identifying vulnerabilities, reducing security risks and improving the security posture of information systems.

1.3.2 IT Infrastructure

The company provides IT infrastructure solutions involving servers, network switching, firewalls and storage systems.

1.3.3 Web Hosting and Domain Registration

YanaCorp provides domain registration and web hosting services, including .tz, .co.tz and other domain extensions. Its website describes the company as an official .TZ registrar.

1.3.4 Software Development

The company develops websites, mobile applications, portals, dashboards and backend software designed around organizational requirements.

1.3.5 Business Automation Systems

YanaCorp provides business automation systems including CRM, POS, HRM, document management, healthcare management, analytics and other enterprise systems.

1.3.6 Technical Support

The organization also provides technical support and infrastructure-related services to help organizations maintain their technology environments.

1.3.7 Training and Capacity Building

YanaCorp provides practical ICT and security training covering areas such as systems, networking, cloud, cybersecurity and data protection.

1.4 ORGANIZATIONAL STRUCTURE

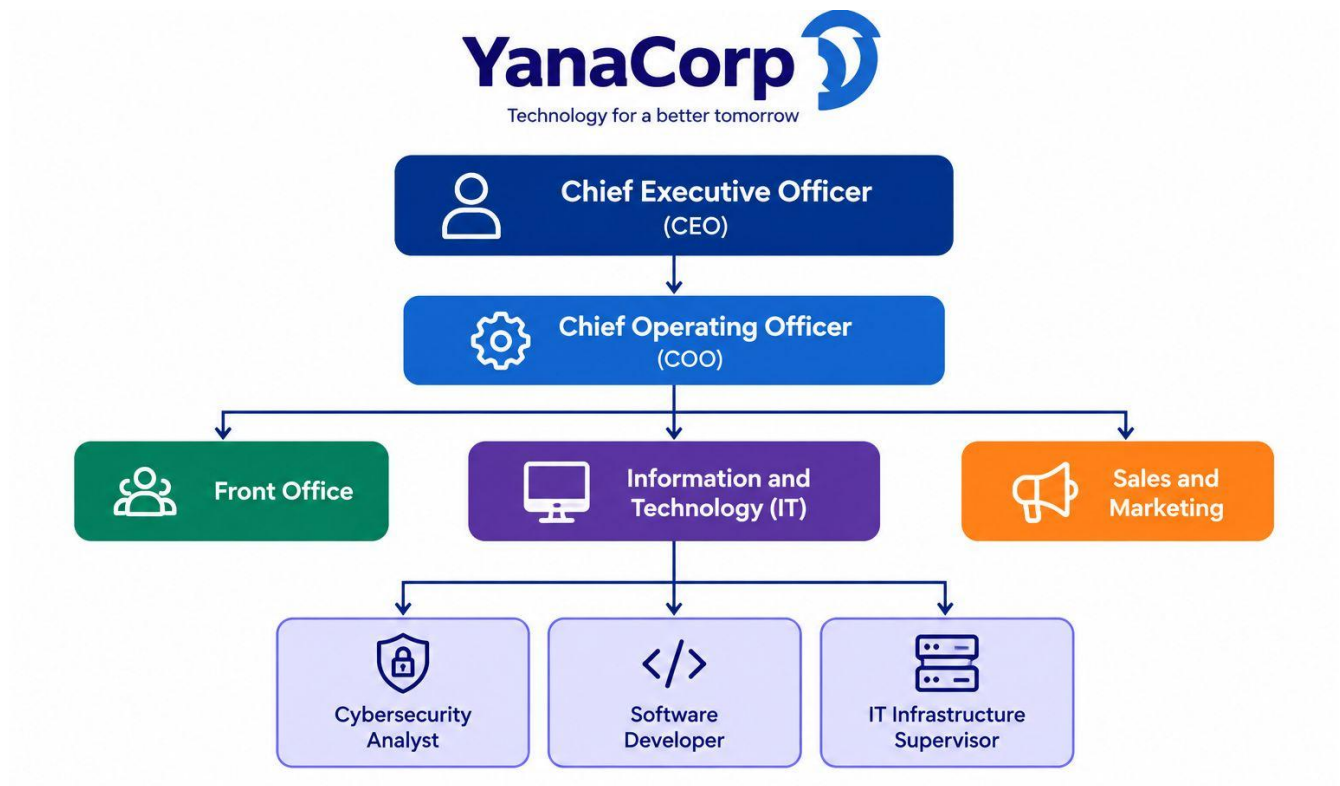


Figure 1:Organizational Structure of YanaCorp

The organizational structure is structured into different key administrative, operational, and technical roles as described below:

1.4.1 Chief Executive Officer (CEO)

The Chief Executive Officer (CEO) is the head of the organization and provides overall leadership and strategic direction. The CEO oversees the general operations and ensures that the organization's objectives are achieved.

1.4.2 Chief Operating Officer (COO)

The Chief Operating Officer (COO) is responsible for coordinating the organization's day-to-day operations. The COO oversees the major operational sections of the organization and ensures that different departments work together effectively.

1.4.3 Front Office

The Front Office handles direct interactions with clients and visitors, including receiving clients, responding to inquiries, coordinating appointments, and directing clients to appropriate departments.

1.4.4 Information and Technology (IT) Department

The IT Department is responsible for the organization's overall technology-related operations and served as the primary placement section during the field attachment. The department comprises the following technical roles:

- **Cybersecurity Analyst:** Responsible for cybersecurity monitoring, vulnerability assessment, security testing, and the protection of information systems.
- **Software Developer:** Responsible for software and system development, maintenance, testing, and implementation.
- **IT Infrastructure Supervisor:** Responsible for coordinating and supervising IT infrastructure, including servers, networks, systems, and technical resources.

1.4.5 Sales and Marketing Department

The Sales and Marketing Department is responsible for promoting the organization's products and services, engaging potential customers, supporting sales activities, and maintaining client relationships.

Overall, this structure enables YanaCorp to coordinate administrative, technical, customer-facing, and business development activities effectively while providing a structured operational environment for cybersecurity and IT tasks.

1.5 NUMBER OF EMPLOYEES

YanaCorp is a technology-driven organization that employs a total of 19 staff members distributed across key functional units, including executive management, the Information and Technology (IT) department, front office administration, and sales and marketing. Personnel from these diverse operational areas collaborate closely, applying their specialized skills to deliver effective ICT solutions and client services. This teamwork involves cybersecurity analysts, software developers, IT infrastructure supervisors, and administrative personnel working collectively to manage network operations, maintain digital systems, ensure system security, and support overall business functions. Through such joint efforts, the technical and administrative teams enable the organization to maintain a secure, reliable, and efficient operating environment.

1.6 ACTIVITIES PERFORMED BY THE ORGANIZATION

YanaCorp performs activities across different areas of ICT, including software development, cybersecurity, IT infrastructure management, hosting, domain registration, technical support and business automation.

During my field attachment, the activities that were most relevant to my Bachelor of Cyber Security programme included practical learning in:

- i. System administration
- ii. Network configuration and troubleshooting
- iii. Linux/Ubuntu Server administration
- iv. Digital certificate configuration
- v. Vulnerability scanning using GVM/OpenVAS
- vi. Endpoint security monitoring and SIEM integration using Wazuh

These activities allowed me to connect theoretical cybersecurity concepts with practical system and network environments.

1.7 LEVEL OF TECHNOLOGY

The organization operates within a technology environment involving modern ICT infrastructure, cybersecurity solutions, servers, networking equipment, software systems and cloud-related technologies.

YanaCorp publicly describes its services as including enterprise infrastructure such as servers, firewalls, core switching networks and storage, together with cybersecurity and software development services.

During my training, I gained practical exposure to both Windows and Linux environments, virtualization and cybersecurity tools. The practical environment included:

- Windows Server 2019
- Ubuntu Server
- Kali Linux
- SSH
- Digital certificates
- Greenbone Vulnerability Management/OpenVAS
- SIEM integration using Wazuh

The use of these technologies enabled me to understand how different components of an IT infrastructure interact with security controls.

1.8 COMPANY CUSTOMERS

YanaCorp serves a diverse client base comprising private-sector businesses, enterprise organizations, financial institutions, educational entities, and public-sector institutions across Tanzania and East Africa. These clients rely on YanaCorp for a wide range of technology solutions, including cybersecurity vulnerability assessments, IT infrastructure deployment, cloud web hosting, domain registration, and custom software development. Additionally, students, early-career technology professionals, and corporate trainees form an essential customer segment that utilizes YanaCorp's specialized technical training and capacity-building programs to enhance their ICT and cybersecurity capabilities.

1.9 STRATEGIC DIRECTION

The strategic direction of YanaCorp focuses on delivering secure, scalable, and innovative technology solutions that empower organizations and businesses across Tanzania and Africa. The company emphasizes driving digital transformation by strengthening IT infrastructure, modernizing enterprise software, and providing robust cybersecurity protections. By integrating security controls, cloud technologies, and technical capacity building, YanaCorp aims to build resilient digital ecosystems that optimize operational performance and protect critical information assets against emerging cyber threats.

1.9.1 Vision:

"To become one of Africa's most trusted technology partners — driving digital transformation, building future-ready systems, and championing women's leadership in technology."

1.9.2 Mission:

"To deliver secure, innovative, and reliable technology solutions that empower institutions, businesses, and communities to operate efficiently, securely, and sustainably."

1.9.3 Values:

- **Innovation:** We foster continuous improvement and creative problem-solving to deliver cutting-edge technology and security solutions tailored to modern digital challenges.
- **Customer-Centricity:** We prioritize our clients' operational goals by providing dependable, custom-tailored ICT services and continuous technical support.

- **Integrity:** We uphold the highest ethical standards, transparency, and confidentiality in handling critical enterprise data and security operations.
- **Collaboration:** We value teamwork and strong strategic partnerships to achieve shared technical excellence and long-term success.
- **Excellence:** We strive for professional superiority in every domain, from software engineering and network infrastructure to cybersecurity monitoring and threat mitigation.

CHAPTER TWO

2.0 ACTIVITIES PERFORMED

2.1 Placement in the Organization and Working Environment

2.1.1 Placement in the Organization

The field attachment was conducted at YanaCorp within the Information and Communications Technology (ICT) Department. The practical training was supervised by Mr. Elisha Alex Mkumbo, who assigned tasks, provided technical guidance, and monitored performance progress throughout the attachment period.

2.1.2 Working Environment

The working environment was supportive, highly collaborative, and well-supervised. It comprised virtualized lab setups (Oracle VirtualBox, Kali Linux, Windows Server 2019, Ubuntu Server 22.04, and Windows 11 clients). This environment provided a safe and controlled platform for executing technical cybersecurity and system administration tasks without affecting live production systems.

2.2 Description of Tasks and Duties Performed

During the attachment, routine technical tasks, problem-solving activities, and practical security assessments were performed:

2.2.1 Windows Server Administration and Active Directory Management

Installed and configured Windows Server 2019 to manage centralized identity, access controls, and domain infrastructure. Active Directory Domain Services (AD DS) was deployed to group users and network resources securely into domain environments. Specific Organizational Units (OUs) were created alongside structured domain user accounts to enforce policy management.

As shown in **Figure 2** below, the Server Manager interface was utilized to monitor local server roles, deployment status, and Active Directory components.

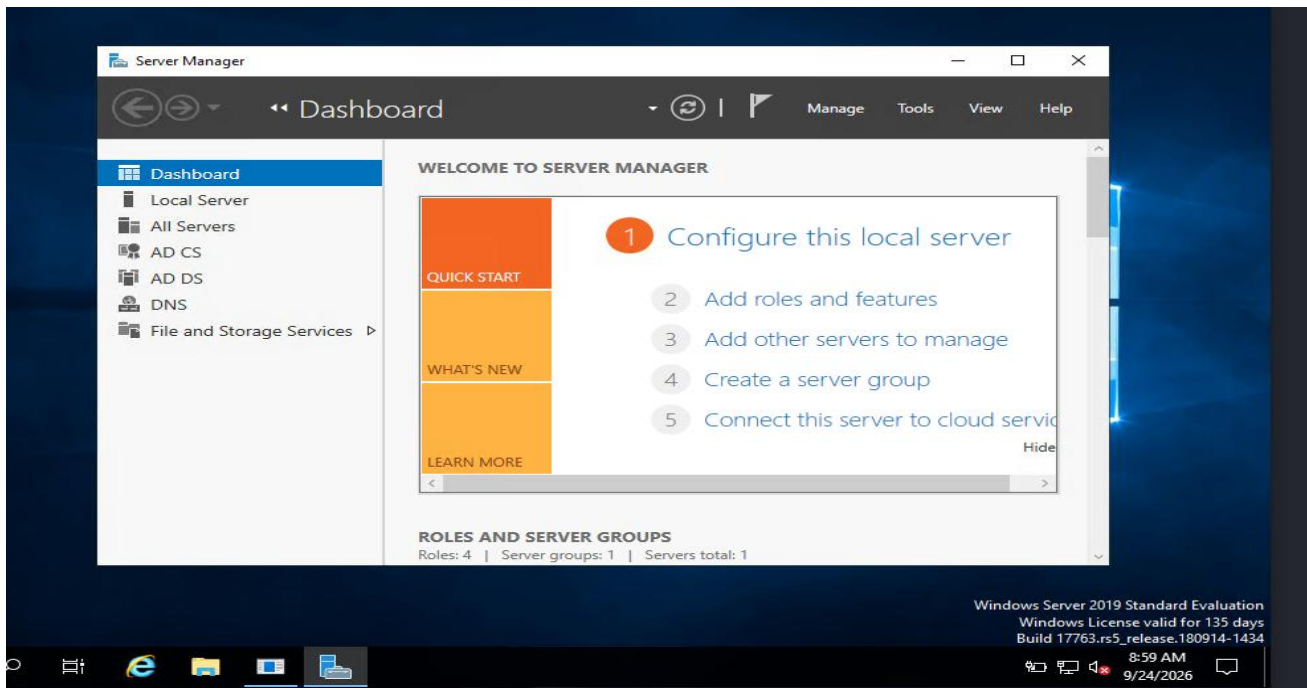


Figure 2: Windows Server 2019 Server Manager

Furthermore, **Figure 3** illustrates the configuration of domain user accounts, administrative groups, and Organizational Units within the Active Directory Users and Computers console.

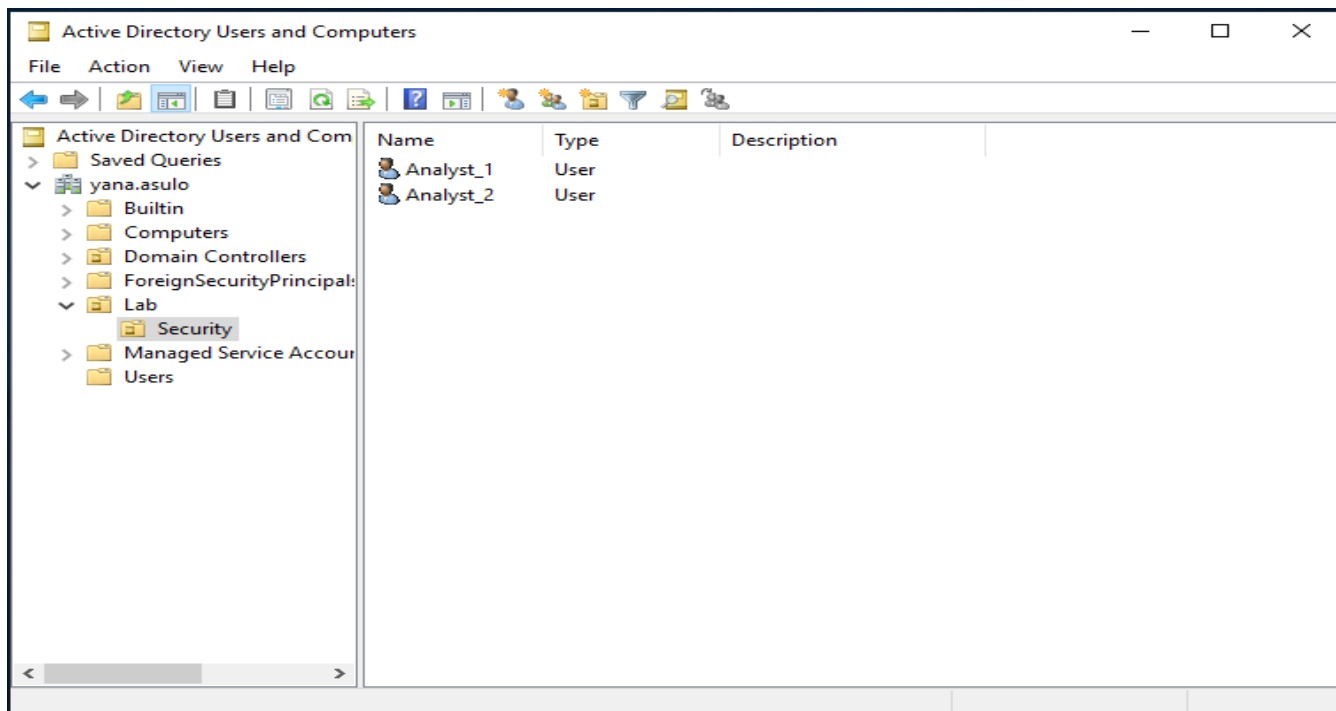


Figure 3: Active Directory Users and Computers Console

2.2.2 Linux Web Server Deployment and Database Configuration

Administered an Ubuntu Server 22.04 LTS instance to deploy web hosting services and backend database solutions. Static IP addressing was configured using Netplan to ensure network stability. An Nginx web server was installed and configured to process incoming web traffic, alongside a MariaDB database server for relational data management.

Figure 4 below displays the Nginx service status, verifying the web server was active, enabled, and functioning correctly on the Ubuntu Linux host.

```
asulo97@ubuntu-server:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-09-24 05:37:50 UTC; 37min ago
     Docs: man:nginx(8)
  Process: 696 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Process: 803 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
 Main PID: 808 (nginx)
    Tasks: 3 (limit: 4553)
   Memory: 12.6M
      CPU: 78ms
  CGroup: /system.slice/nginx.service
          └─808 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─809 "nginx: worker process"
                └─810 "nginx: worker process"

Sep 24 05:37:50 ubuntu-server systemd[1]: Starting A high performance web server: nginx.
Sep 24 05:37:50 ubuntu-server systemd[1]: Started A high performance web server: nginx.
lines 1-17/17 (END)...skipping...
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-09-24 05:37:50 UTC; 37min ago
     Docs: man:nginx(8)
  Process: 696 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Process: 803 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
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             └─809 "nginx: worker process"
                └─810 "nginx: worker process"
```

Figure 4: Nginx Web Server Active Status on Ubuntu Server

2.2.3 Digital Certificate Management and Public Key Infrastructure (PKI)

Configured Active Directory Certificate Services (AD CS) to function as a Certificate Authority (CA) within the enterprise environment. This setup facilitated the generation, enrollment, and validation of digital certificates required for securing web traffic (TLS/SSL) and authenticating internal domain services.

As depicted in **Figure 5**, the Certification Authority management console was configured to handle public key infrastructure services.

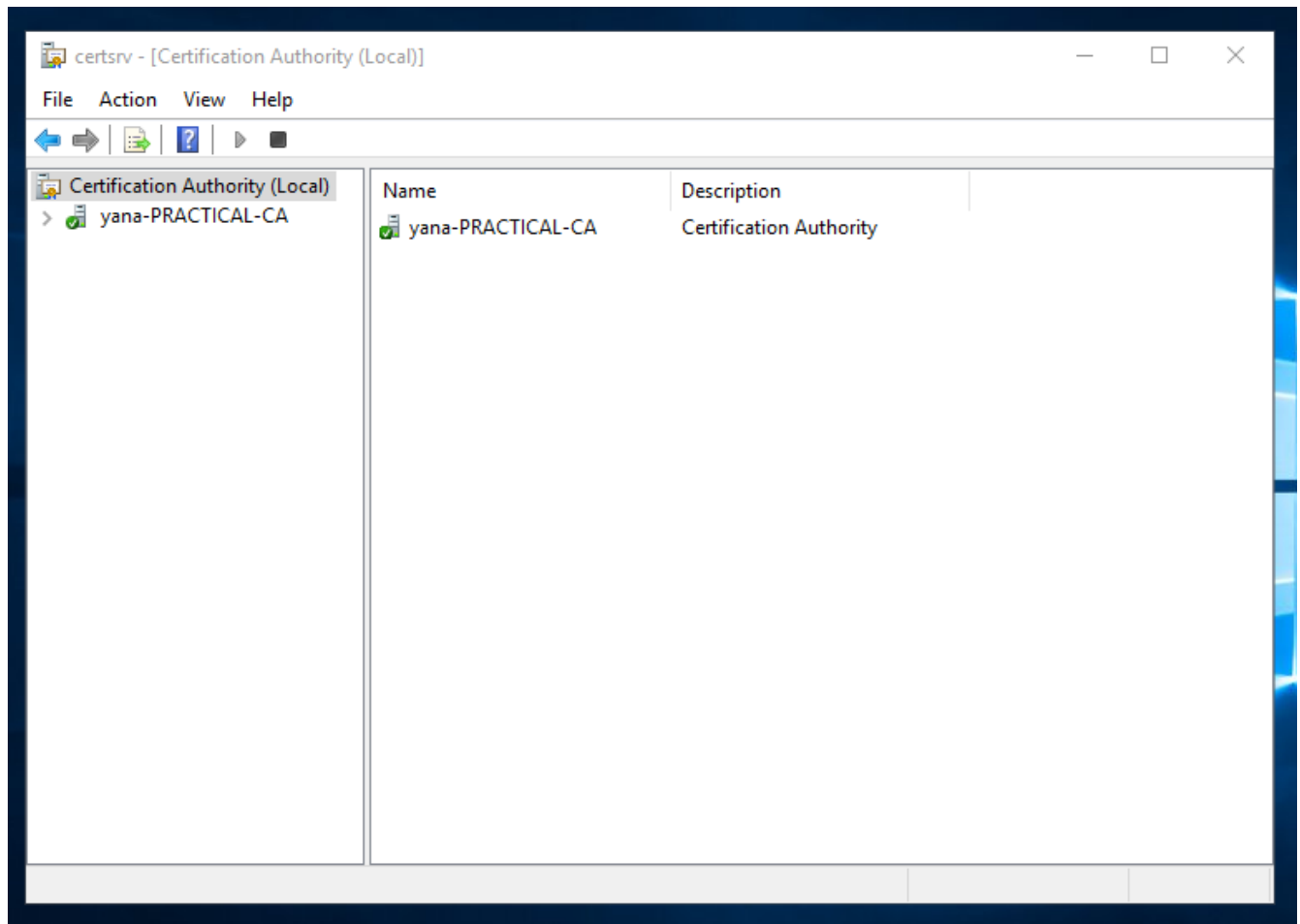


Figure 5: Enterprise Certification Authority Console

2.2.4 Vulnerability Management & System Hardening

Installed Greenbone Vulnerability Management (GVM/OpenVAS) on Kali Linux, conducted authenticated/unauthenticated scans on Linux and Windows targets, remediated identified flaws (SSH hardening, microcode updates, TLS configuration), and verified clean re-scans.

As illustrated in Figure 6 below, automated vulnerability scans were executed targeting Windows and Ubuntu hosts.

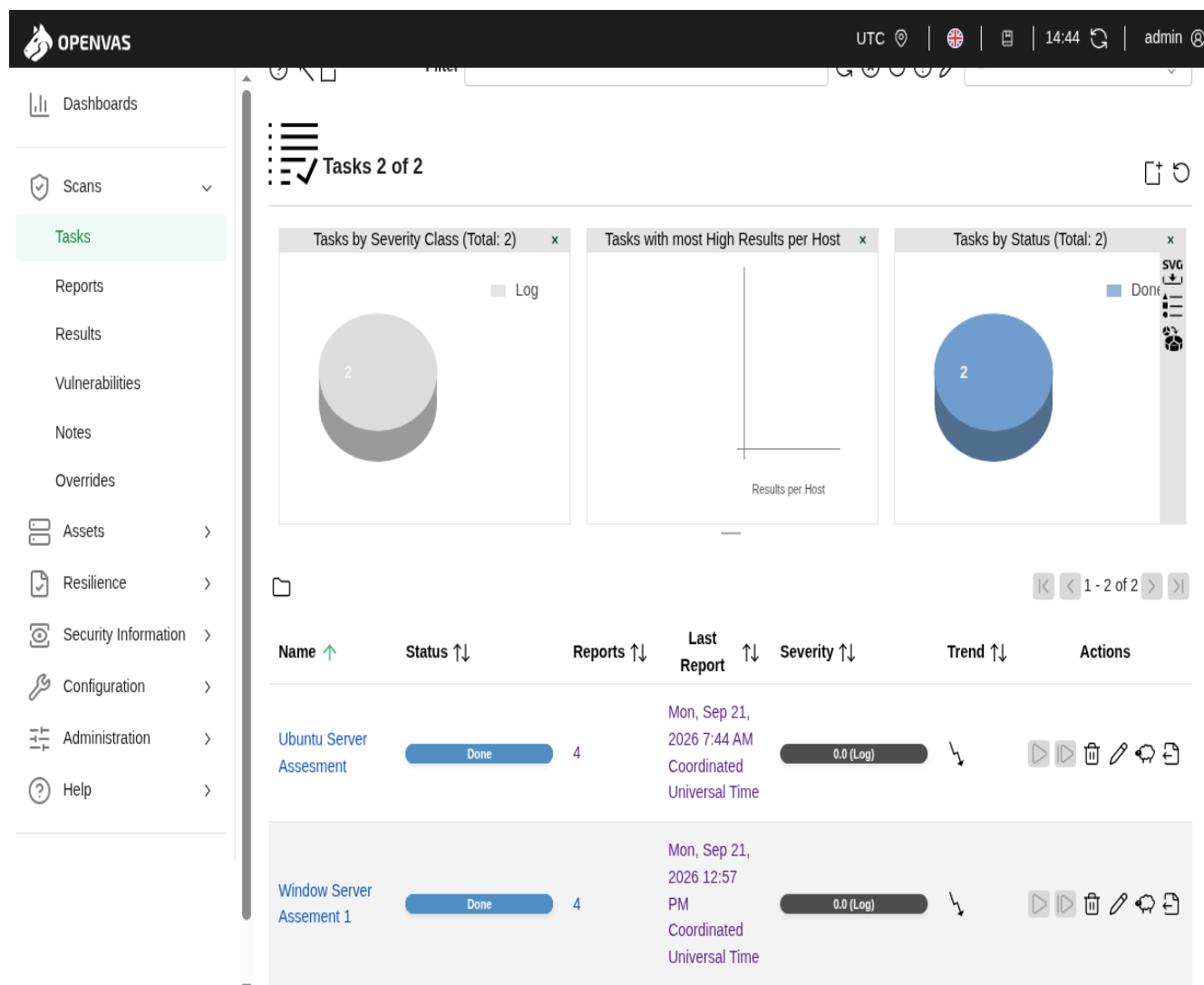


Figure 6: GVM Dashboard / Vulnerability Scan Results

2.2.5 Security Monitoring & Endpoint Security (Wazuh)

Administered Wazuh endpoint agents on Windows workstations, performed agent upgrades, re-authenticated endpoints to wazuh, coordinated with the Infrastructure Supervisor for firewall permissions, and verified SOC integration workflows.

Figure 7 depicts the active Wazuh endpoint security dashboard showing real-time agent telemetry.

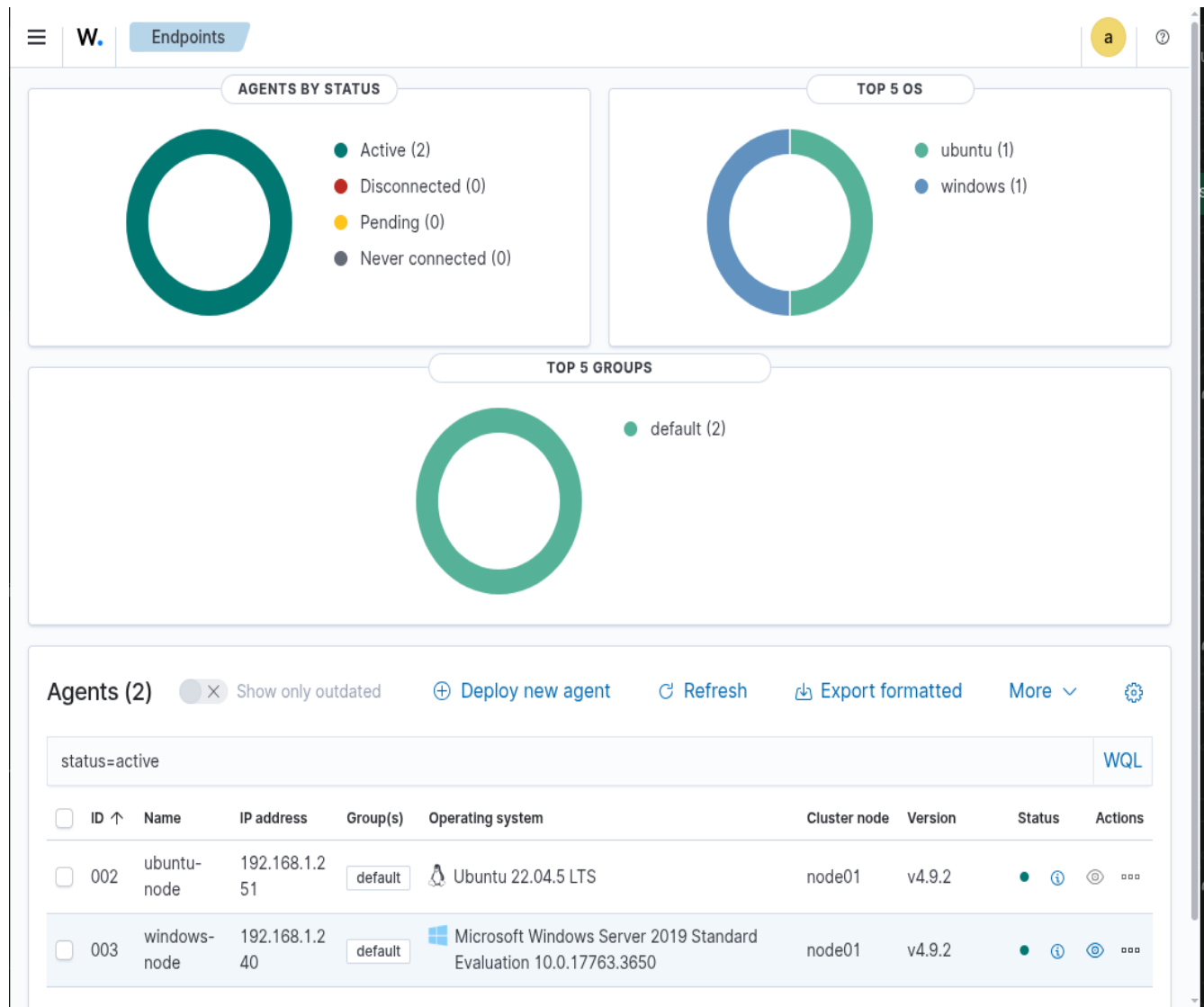


Figure 7: Wazuh Security Dashboard and Endpoints

2.3 Feelings Towards Tasks Performed

The assigned technical duties were extremely engaging, practical, and highly relevant to modern cybersecurity operations. Configuring enterprise-grade platforms such as Wazuh SIEM, Active Directory Domain Services, and Greenbone Vulnerability Management (GVM) provided immense professional satisfaction. The hands-on exposure converted theoretical cybersecurity concepts into practical infrastructure management, driving strong motivation to solve complex technical problems.

2.4 Demands of Assigned Tasks and Personal Performance

The demands of the assigned tasks were technically rigorous and required analytical thinking, system-level troubleshooting, and creative problem-solving. Tasks such as configuring static network interfaces, diagnosing database connection errors during GVM setup, and managing privilege escalation requirements demanded high attention to detail. Personal performance successfully met these operational demands through persistent technical research, logical diagnostic workflows, and effective collaboration with the IT infrastructure team.

2.5 Problems Encountered and Solutions Applied

2.5.1 Technical Problems Encountered

During the execution of assigned duties, several technical challenges were encountered:

- i. **Netplan Configuration Errors:** Syntax and formatting errors in Netplan YAML files on Ubuntu Server causing loss of network connectivity.
- ii. **Privilege Escalation Issues in GVM:** Authenticated vulnerability scanning failures on Linux target systems due to insufficient SSH privileges.
- iii. **Wazuh Agent Authentication & Firewall Blocks:** Communication failures between remote endpoints and the Wazuh manager caused by active firewall restrictions and legacy agent tokens.
- iv. **Tool Limitations in Burp Suite Community:** Restrictions on automated web security testing and out-of-band interactions during SSRF assessment.

2.5.2 Solutions Applied and Recommendations

- **Solutions Applied:** Netplan YAML configurations were validated using Linux CLI utilities (netplan try); /etc/sudoers permissions were updated via visudo (NOPASSWD: ALL) for the GVM service account; and network firewall rules were adjusted in coordination with the IT Infrastructure Supervisor to permit Wazuh agent traffic on port 1514/1515.
- **Recommendations:** Academic institutions should provide students with dedicated cloud or virtualized lab environments equipped with licensed/full-featured security testing tools. Additionally, host organizations should maintain documented network policy exceptions for training subnetworks.

2.6 Expectations and Learning Outcomes

2.6.1 Expectations Before Training

Prior to joining YanaCorp, the primary expectation was to transition from theoretical cybersecurity knowledge to real-world, hands-on operational practice. This expectation was fully realized through active involvement in domain administration, endpoint monitoring, web application security testing, and vulnerability management.

2.6.2 New Skills and Knowledge Acquired

The field attachment enabled the acquisition of critical technical competencies, including:

- i. Deploying and managing Wazuh agents for centralized security monitoring and SIEM log analysis.
- ii. Building Active Directory structures, configuring Organizational Units (OUs), and managing domain access controls in Windows Server 2019.
- iii. Conducting authenticated vulnerability scans, interpreting CVSS severity metrics, and performing system hardening using OpenVAS/GVM.
- iv. Intercepting, analyzing, and manipulating HTTP traffic using Burp Suite and FoxyProxy to identify web application vulnerabilities.

2.7 Gaps Between Theory and Practice

A noticeable gap exists between classroom theoretical models and enterprise operational reality. Academic coursework primarily focuses on idealized configurations and isolated security concepts. In practice, enterprise environments involve complex dependencies, legacy system compatibility issues, strict production uptime constraints, and multi-layered network security policies that complicate standard deployments (such as agent installation and automated scanning).

2.8 Employment Prospects

- **Desire for Employment:** Working at YanaCorp as a Cybersecurity Analyst or System Administrator would be highly desirable due to the organization's dynamic technical environment and focus on modern ICT solutions.
- **Employer Feedback & Prospects:** Based on technical performance, adaptability, and successful task execution during the attachment, the supervisors expressed strong confidence in my technical capabilities.
- **Employment Promises:** While no formal employment contract was issued due to policy requirements, positive feedback was provided regarding future entry-level opportunities upon completion of degree studies.

2.9 Relevancy of the Attachment

The field attachment was highly relevant to the Bachelor of Science in Cybersecurity degree program. It directly reinforced core curriculum areas such as Network Security, System Administration, Penetration Testing, and Vulnerability Assessment. The experience effectively bridged the gap between academic theory and practical execution, fostering professional confidence, technical competence, and industry-standard operational habits.

CHAPTER THREE

3.0 CONCLUSION AND RECOMMENDATIONS

3.1 Conclusion

The field attachment undertaken at YanaCorp within the Information and Communications Technology (ICT) Department provided invaluable practical exposure to core cybersecurity, network infrastructure, and system administration practices. Active involvement in server management, endpoint security monitoring, web application testing, and vulnerability management successfully bridged the gap between academic theory and enterprise operations.

3.1.1 Major Strengths of the Field Attachment Programme

The field attachment programme exhibited several key strengths:

- i. **Hands-on Exposure to Enterprise Security Tools:** Direct practical engagement with industry-standard technologies including Active Directory, Wazuh SIEM, GVM/OpenVAS, Burp Suite, and Kali Linux.
- ii. **Comprehensive Vulnerability Management Lifecycle:** Emphasis on the end-to-end security cycle (Identification → Risk Analysis → Remediation → Verification Scanning) rather than basic scanning.
- iii. **Realistic Operational Problem-Solving:** Diagnostic exposure to live technical errors (e.g., Netplan configuration issues, database service errors, and SSH privilege escalation blocks) enhanced troubleshooting capabilities.
- iv. **Multi-Domain Skill Integration:** Balanced exposure across system administration, identity and access management, endpoint monitoring, web security, and SOC alert workflows.

3.1.2 Major Weaknesses of the Field Attachment Programme

Despite its success, the attachment experienced certain limitations:

- i. **Time Constraints vs. Technical Scope:** The attachment duration was insufficient to fully master advanced SOC orchestration workflows (Wazuh, TheHive, Cortex) and incident response frameworks.

- ii. Software Licensing and Tool Restrictions: Security testing was constrained by community-edition tool limitations (e.g., absence of Burp Collaborator for Blind SSRF) and restricted privileges during domain scanning.
- iii. Virtualized Lab Boundaries: Security exercises were primarily restricted to virtualized lab environments, which do not fully simulate the operational scale, latency, and complexity of live production networks.

3.1.3 Benefits Derived from the Field Attachment Programme

The practical training yielded significant technical and professional benefits:

- i. Technical Competence: Acquired hands-on capability in configuring Wazuh agents, managing Active Directory OUs and users, analyzing HTTP traffic, and executing automated vulnerability scans.
- ii. Diagnostic and Analytical Skills: Developed a structured methodology for troubleshooting server network errors, database dependencies, and Linux security configurations.
- iii. Professional and Soft Skills Development: Improved technical documentation, formal reporting under supervision, time management, and adherence to enterprise cybersecurity ethics.

3.2 Recommendations

3.2.1 Recommendations for Improving Gaps Between Theory and Practice

- i. Expanded Academic Hands-On Labs: Universities should incorporate end-to-end practical labs focusing on vulnerability remediation and server hardening before students proceed to field attachments.
- ii. Multi-Domain Capstone Projects: Curricula should feature integrated projects combining Active Directory, SIEM monitoring, and web application security to reflect real-world enterprise architectures.
- iii. Practical Offensive and Defensive Exercises: Academic training should balance vulnerability exploitation tools (Burp Suite, Metasploit) with defensive controls and system hardening protocols.

3.2.2 Recommendations for Planning and Operational Logistics

- i. Pre-Attachment Technical Alignment: Universities should provide host organizations with clear skill profiles, and organizations should prepare required virtual lab resources prior to student arrival.
- ii. Structured Mentorship Reviews: University supervisors and host mentors should establish bi-weekly progress evaluations to systematically review student learning milestones and technical challenges.
- iii. Dedicated Training Subnets: Host organizations should maintain dedicated, policy-exempt training subnets allowing students to safely conduct vulnerability scans and agent deployments without firewall interference.

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APPENDICES

APPENDIX 1

- Field Attachment Logbook
- Arrival Note (Form No. 2)
- On-Site Supervisor's Assessment Form (Form No. 1)
- Certificate of Successful Completion (Form No. 6)

APPENDIX 2

- Field Placement Confirmation Letter
- Sample Vulnerability Scan Assessment Report